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ITA0502-COMPUTER VISION

Activity-1

1. Medical Image Enhancement

A doctor receives an X-ray image in which the bones are visible, but the darker tissues have very poor visibility.

Questions:

1. Which gray-level transformation would you recommend?
2. Why is it suitable?
3. What effect will it have on the image?

2. Night-Time Surveillance

A CCTV camera captures a parking lot at night. Most pixel values are concentrated between 0 and 40, making the image very dark.

Questions:

1. Which transformation would improve the visibility?
2. Why is it more suitable than negative transformation?
3. Predict the appearance of the enhanced image.

3. Fingerprint Recognition

A fingerprint image contains many fine ridge details, but the contrast between the ridges and valleys is very low.

Questions:

1. Which gray-level transformation would you apply?
2. Explain why.
3. What would happen if a log transformation were applied instead?

Medical Image Enhancement (X-ray):

1. Recommended transformation:- Contrast stretching, possibly combined with a power-law (Gamma) transformation with $\gamma=1$.

2. Why suitable:- X-rays have bones and soft tissue crammed into a narrow gray-level range. Contrast stretching expands this narrow range to cover the full dynamic range, spreading out the poorly visible dark tissue pixels so they become distinguishable.

3. Effect on image:

The dark tissue regions become brighter and more distinguishable, contrast between the tissues and bones improves, and previously indistinguishable details in shadow regions become visible — without saturating the already-bright bone areas.

2. Night-Time Surveillance (CCTV) :

1. Recommended transformation:-

Power-Law (Gamma) transformation with $\gamma < 1$, or Histogram Equalization.

2. Why better than negative transformation:-

Negative transformation ($s = L - 1 - r$) simply inverts intensities - it would turn the dark background bright and vice versa, which does not improve visibility, it's just flip dark and light regions. Gamma correction ($\gamma < 1$) or Histogram equalization instead redistributes/expands the narrow low-intensity ~~red~~ range (0-40) across the full range.

3. predicted appearance:

The image becomes noticeably brighter overall, previously hidden objects/people in the dark parking lot become visible, and the contrast b/w subtle intensity variations in the 0-40 range is enhanced - without inverting colors.

5. Fingerprint Recognition:

1. Recommended transformation:

Contrast stretching - a linear / piecewise-linear transformation that maximizes contrast between ridges and valleys.

2. Explanation: Fingerprint ridges and valleys occupy a very narrow, closely-space gray-level band. Contrast stretching linearly expands this narrow band across the entire available intensity range, sharply increasing the distinguishability between ridge & valley pixels.

3. If log transformation applied instead:

Log transformation ($S = C \cdot \log(1+r)$) compresses the higher intensity range and expands only the lower (dark) intensity range. Since ridge-valley difference aren't confined to just the dark end of the spectrum, this would not adequately separate ridges from valleys - it might even further compress meaningful mid-to-high range differences.

Resulting in reduced contrast and loss of fine ridge detail, making the fingerprint the harder to recognize accurately.